

# ECON 640: Advanced Macroeconomics and DSGE Modeling

Term: Fall 2026 | Instructor: Dr. Robert Lucas (rlucas@econ.univ.edu)

## Course Description & Overview:

Dynamic stochastic general equilibrium (DSGE) macroeconomic models, New Keynesian staggered pricing, and monetary policy frontiers.

## COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

| Assignment Description                      | Weight | Due Module | Deliverable Format                  |
|---------------------------------------------|--------|------------|-------------------------------------|
| Real Business Cycle (RBC) Dynare Simulation | 30%    | Week 05    | Dynare Mod File & Impulse Responses |
| New Keynesian Monetary Policy Paper         | 30%    | Week 09    | Theoretical Paper & Proofs          |
| Empirical Macroeconometric Term Project     | 30%    | Week 12    | Empirical Estimation Paper          |
| Macro Research Seminar Presentations        | 10%    | Continuous | Discussion Participation            |

## WEEKLY TERM SCHEDULE & READING REQUIREMENTS

| Timeline | Technical Topic Focus                             | Assigned Readings & Docs                   |
|----------|---------------------------------------------------|--------------------------------------------|
| Week 01  | Module 01: Dynamic Programming & Bellman Equation | Stokey, Lucas & Prescott (Ch. 1–4)         |
| Week 02  | Module 02: Ramsey-Cass-Koopmans Growth Model      | Romer (Ch. 2)                              |
| Week 03  | Module 03: Overlapping Generations (OLG) Models   | Diamond (1965); Romer (Ch. 3)              |
| Week 04  | Module 04: Real Business Cycle (RBC) Theory       | Kydland & Prescott (1982)                  |
| Week 05  | Module 05: Solving DSGE Models via Dynare         | Blanchard & Kahn (1980); Dynare User Guide |
| Week 06  | Module 06: Nominal Rigidities & Calvo Pricing     | Calvo (1983)                               |
| Week 07  | Module 07: The Canonical New Keynesian Model      | Gali (Ch. 1–3); Woodford (Ch. 2)           |
| Week 08  | Module 08: Optimal Monetary Policy & Taylor Rule  | Taylor (1993); Clarida et al. (1999)       |
| Week 09  | Module 09: The Zero Lower Bound & QE Policies     | Eggertsson & Woodford (2003)               |
| Week 10  | Module 10: Fiscal Multipliers & Sovereign Debt    | Ramey (2011); Reinhart & Rogoff (2010)     |
| Week 11  | Module 11: Financial Frictions & Credit Cycles    | Bernanke, Gertler & Gilchrist (1999)       |
| Week 12  | Module 12: Term Project Presentations & Review    | Macroeconomics Research Review             |